Lab 2: Music Genre Classification – CNNs

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Google Colab: https://colab.research.google.com/drive/19NLrd5VGHWgPSrdbbxzbx4UzCCs9xQTr?usp=sharing

Setup

We use the GTZAN mel spectrograms (999 songs) with one stratified 80/20 split, shared by every model (200 test songs). To choose configurations, the 80% training set is split again into train and validation, so the test set is only used at the end. Three approaches are compared: a CNN trained from scratch, ResNet18 pretrained on ImageNet (frozen + logistic regression, and full fine-tuning), and the Lab 1 models retrained on this same split.

Results

Task 1. The CNN stacks Conv 3×3 + BatchNorm + MaxPool blocks and ends with global average pooling. Two layers are not enough, and accuracy stops improving after four. The best configuration on validation has 4 layers and 16 filters.

Conv layers	2	3	4	5
16 filters	0.512	0.688	**0.788**	0.762
32 filters	0.556	0.731	0.781	0.775

Validation accuracy of each CNN configuration.

Task 2 and comparison. The table shows the accuracy on the shared 200-song test set and, for the four best models, the mean over 5 different splits. In Lab 1 the Random Forest reached 0.794 in cross-validation; here it drops to 0.660, so that figure was optimistic.

Model	Test acc	Mean over 5 splits	Parameters
ResNet18 fine-tuning	0.810	0.730 ± 0.037	11.2M
CNN from scratch (4 layers / 16 filters)	0.765	0.725 ± 0.009	99k
ANN – descriptors (Lab 1)	0.755	0.745 ± 0.045	-
ResNet18 frozen + LogReg	0.745	0.682 ± 0.019	11.2M
Logistic Regression (Lab 1)	0.690	-	-
Random Forest (Lab 1)	0.660	-	-
k-NN (Lab 1)	0.630	-	-
Decision Tree (Lab 1)	0.495	-	-

Results of all models.

Research questions

Q1. Do learned features beat hand-crafted descriptors?

Not clearly. On the main split, the fine-tuned ResNet18 (0.810) beats the best descriptor model, the ANN (0.755), by 11 songs, but the CNN from scratch (0.765) wins by only 2 songs. Over 5 splits the ANN has the highest mean (0.745, against 0.730 and 0.725), and the gaps are within the noise. Learned features also need to be learned for this task: frozen ImageNet embeddings stay below the descriptors.

Q2. Do CNNs and descriptor models make the same mistakes?

Only partly. All three models confuse country ↔ rock, and rock is one of the hardest genres for all of them (10-13 out of 20). This looks like a property of the music, since these genres share instrumentation. The other errors depend on the model. The CNNs mostly confuse pop ↔ rock. The ANN confuses disco ↔ hiphop and pop ↔ reggae, genres with similar rhythm and energy. ResNet18 and the ANN are both wrong on only 19 songs. If we could pick the right model for each song, accuracy would reach 0.905, so combining them should help.

Q5. How much does one 80/20 split tell you?

Very little. Across 5 splits, the ANN ranges from 0.690 to 0.795 (21 songs) and the fine-tuned ResNet18 from 0.665 to 0.755. These ranges are larger than the gaps between models on a single split. The ranking even changes: on the main split fine-tuning is the best model, but it beats the ANN on only 1 of the 5 splits. The only result that holds on every split is that the frozen embeddings are worse (0.682).